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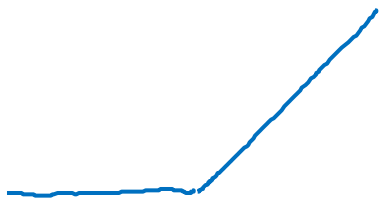
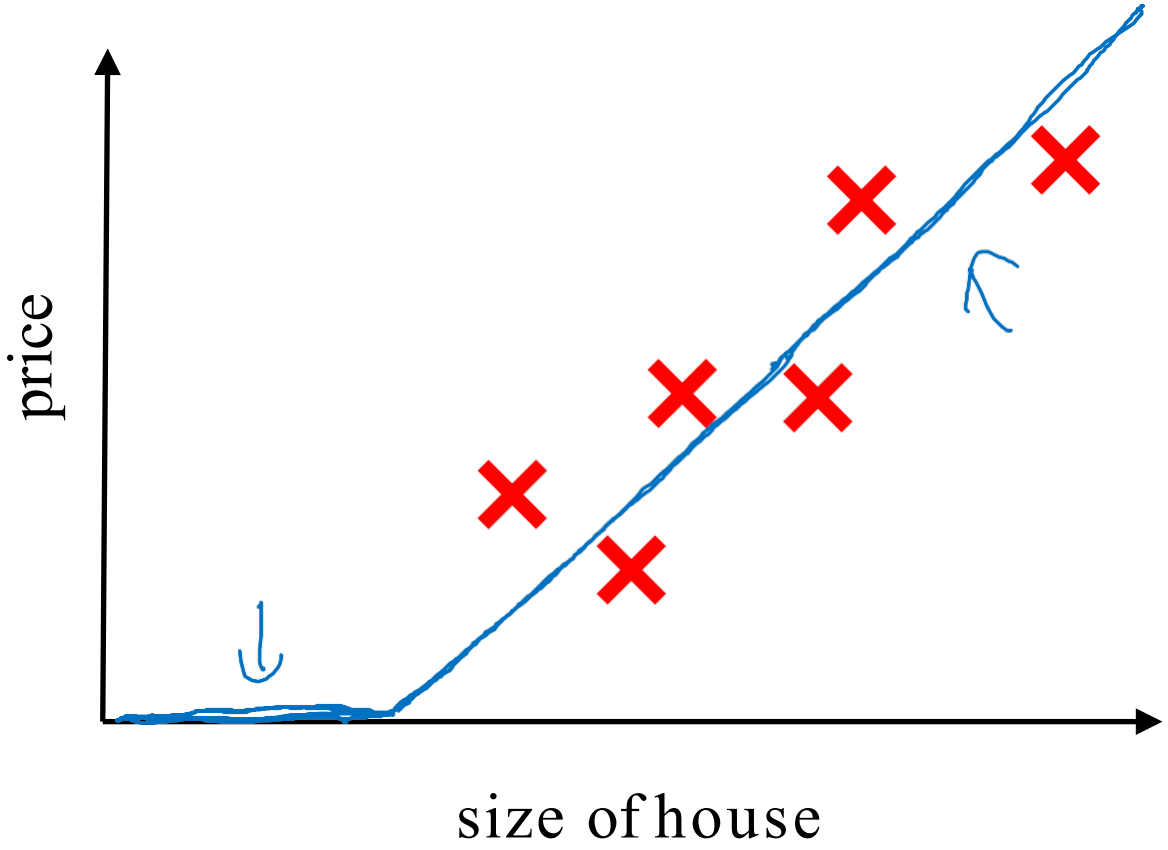
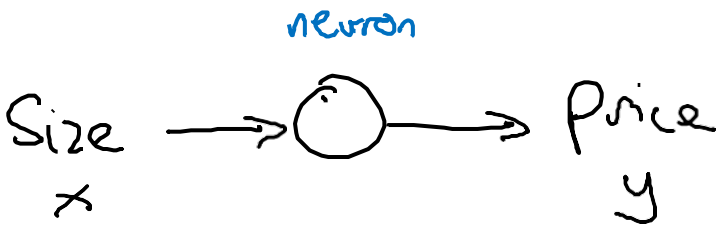
What is a Neural Network?

Housing Price Prediction

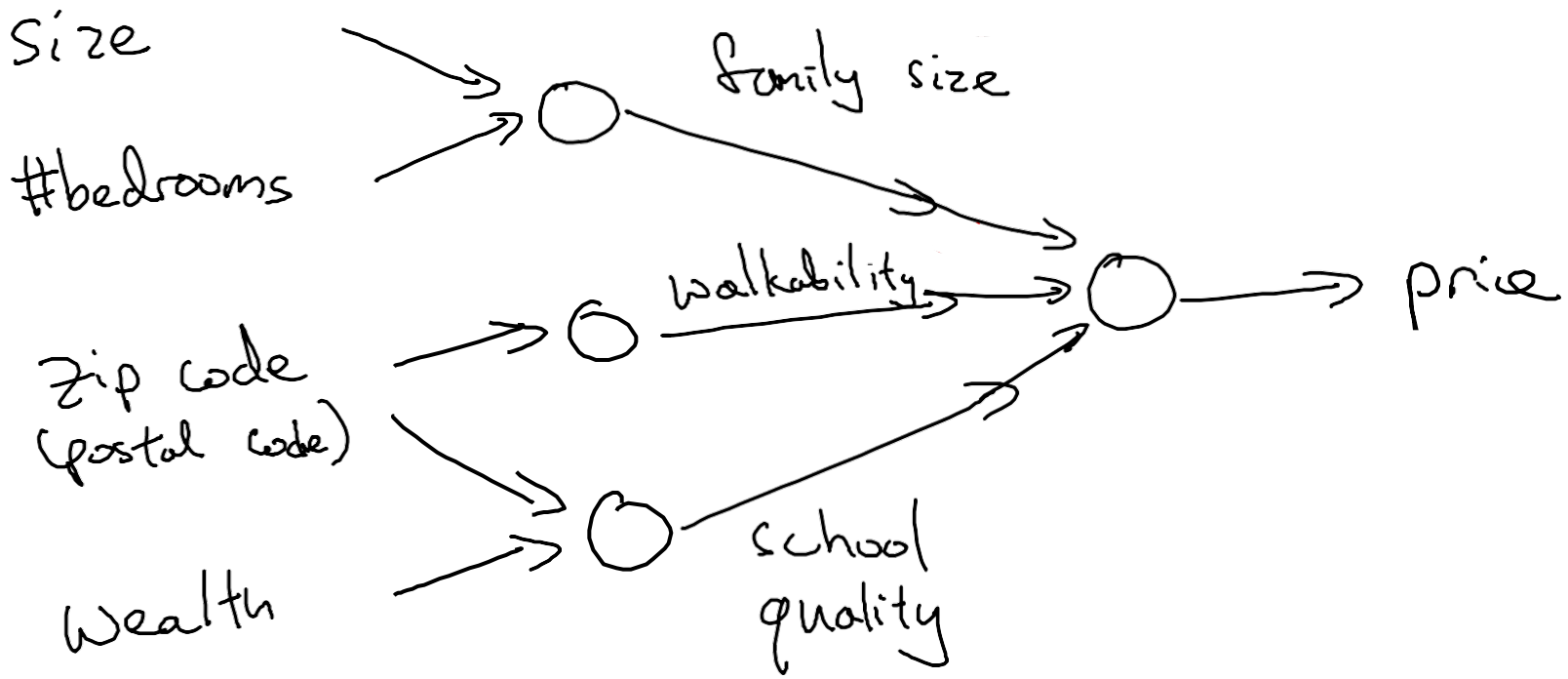


ReLU
Rectified
Linear
Unit

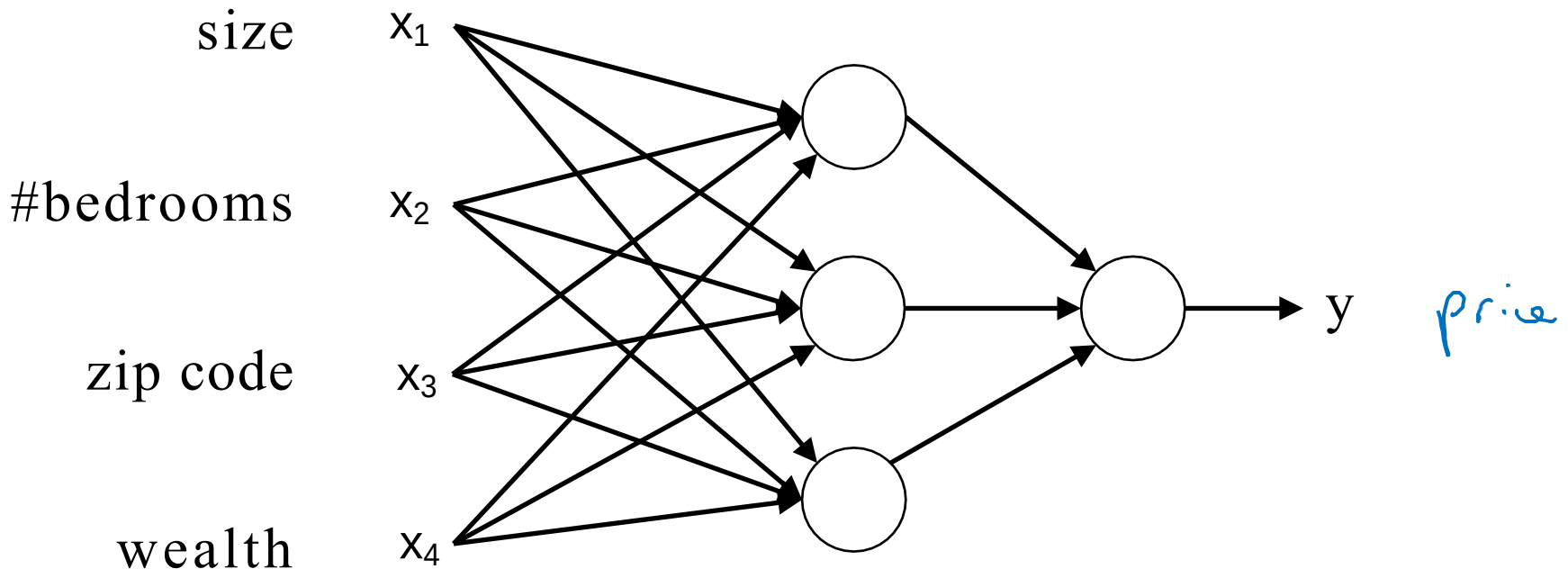




Housing Price Prediction



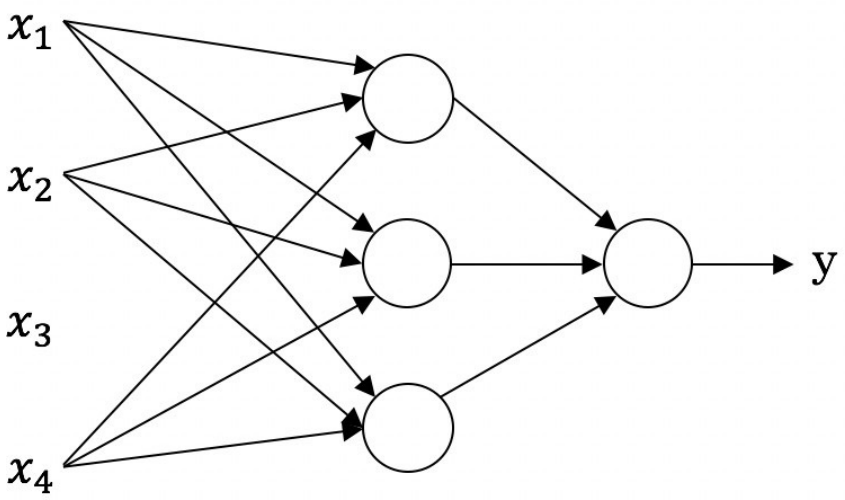
Housing Price Prediction

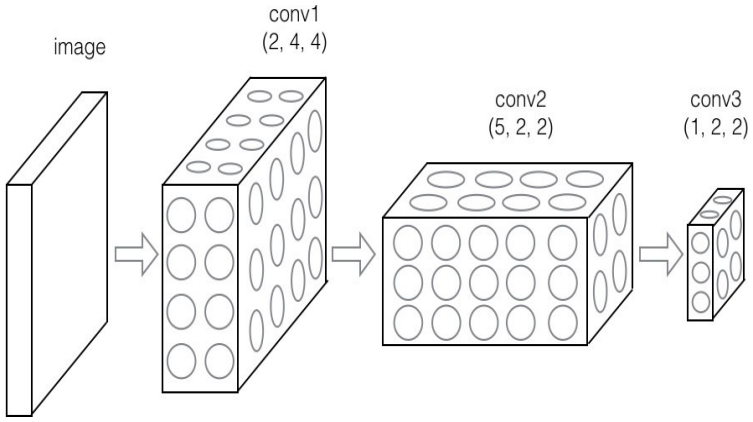


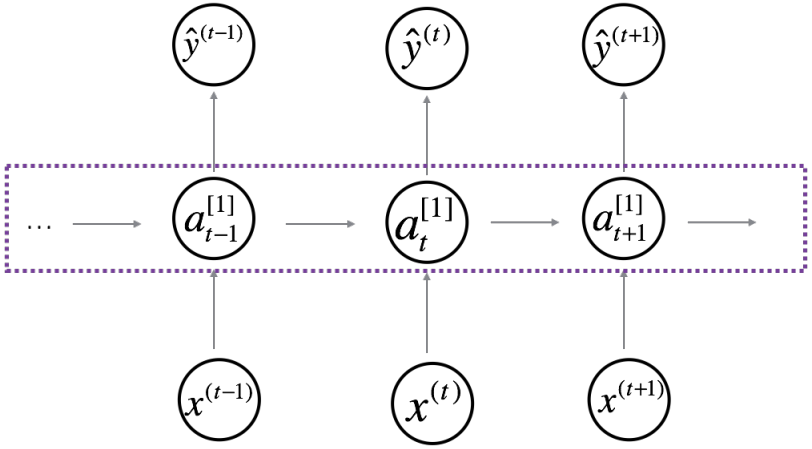
Supervised Learning

Input(x)	Output (y)	Application	
Home features	Price	Real Estate	Standard NN
Ad, user info	Click on ad? (0/1)	Online Advertising	
Image	Object (1,…,1000)	Photo tagging	CNN
Audio	Text transcript	Speech recognition	RNN
English	Chinese	Machine translation	
Image, Radar info	Position of other cars	Autonomous driving	Custom Hybrid

Neural Network examples







Standard NN

Convolutional NN

Recurrent NN

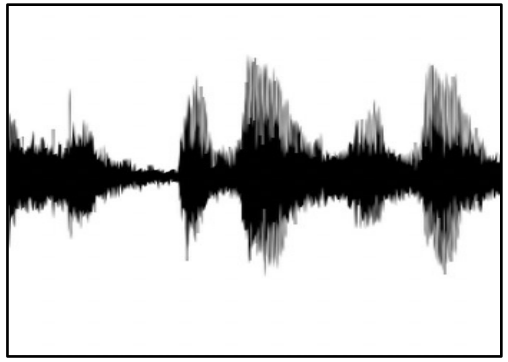
Supervised Learning

Structured Data

Size	#bedrooms	…	Price (1000$s)
2104	3		400
1600	3		330
2400	3		369
…	…		…
3000	4		540

User Age	Ad Id	…	Click
41	93242		1
80	93287		0
18	87312		1
…	…		…
27	71244		1

Unstructured Data



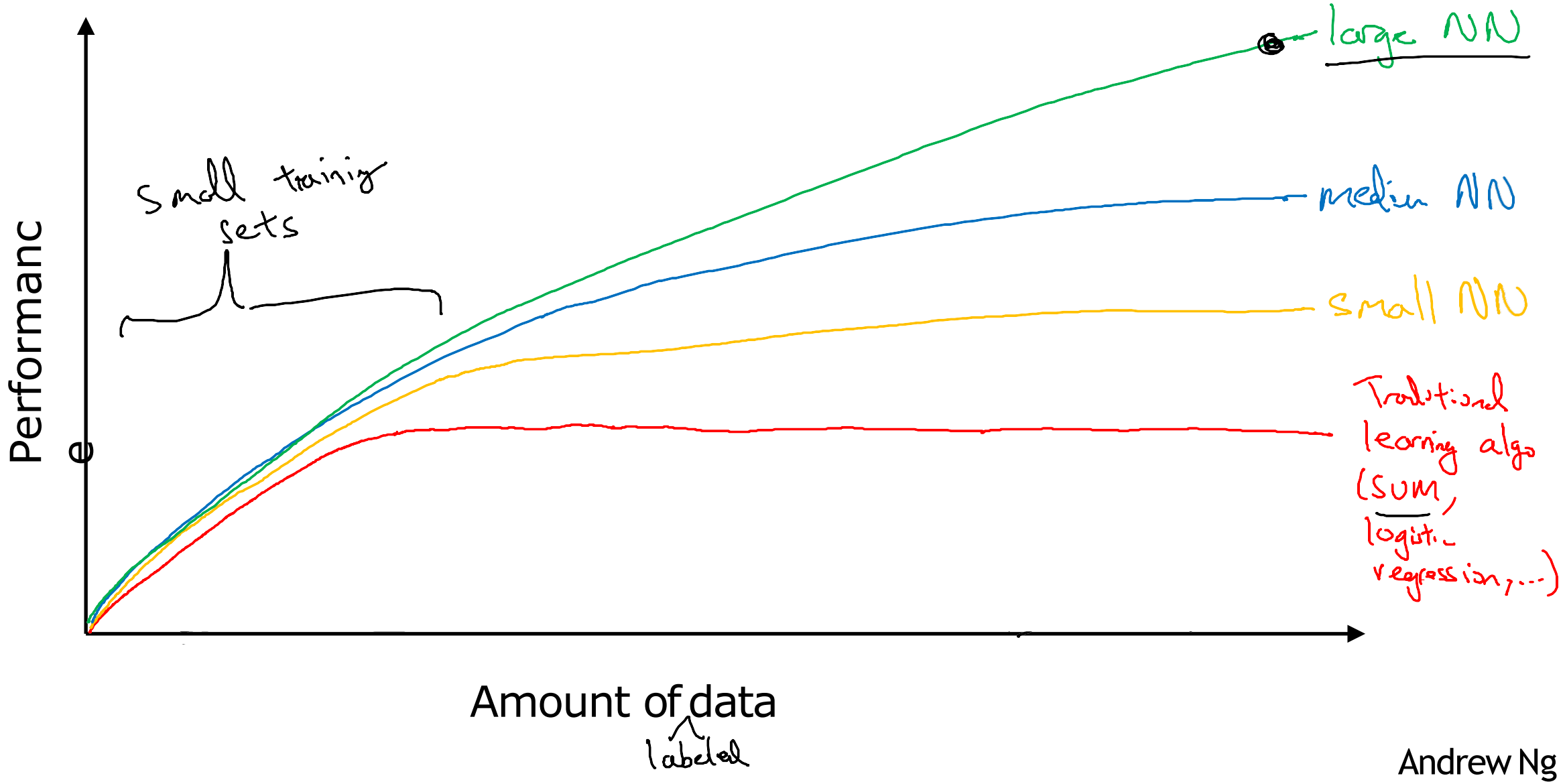
Audio

Image

Four scores and seven years ago…

Text

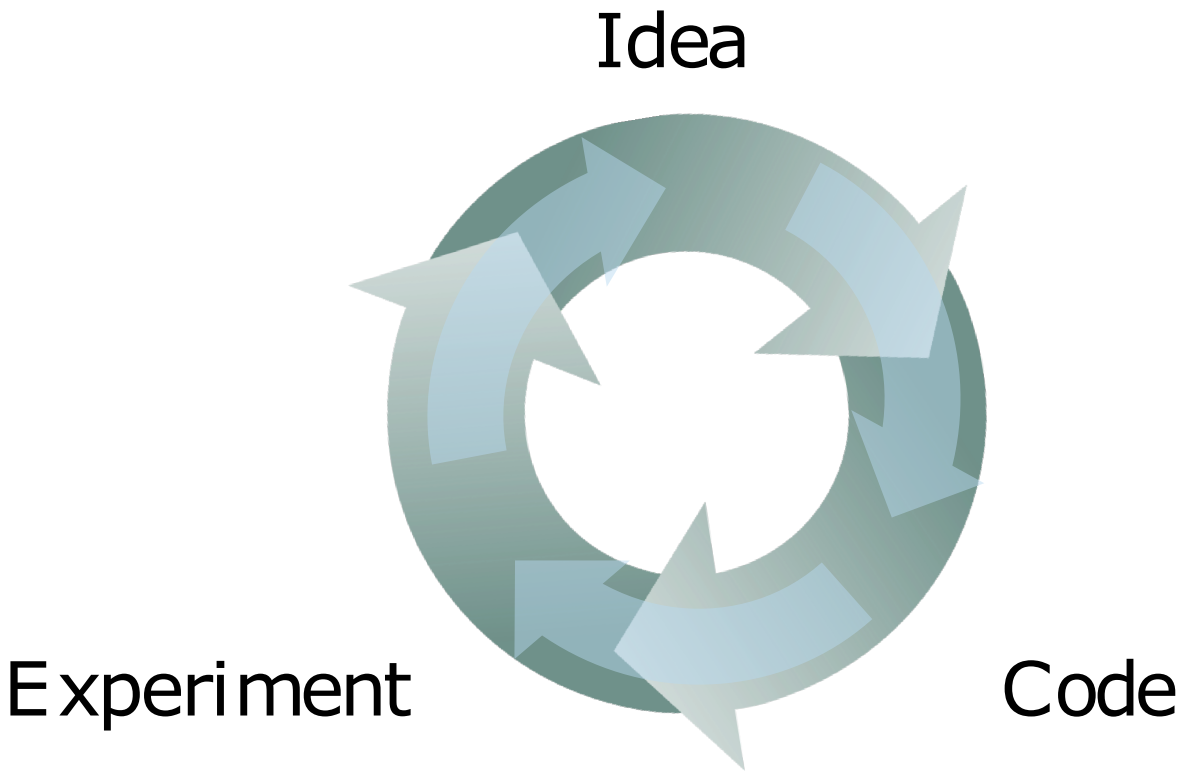
Scale drives deep learning progress



Andrew Ng

Scale drives deep learning progress

- Data
- Computation
- Algorithms



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